

Model predictive control based path following for a wheel-legged robot

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Abstract:

This paper proposes a model predictive control (MPC) based path following approach for tracking control of a wheel-legged robot named BIT-NAZA. The accuracy and stability of tracking control are still the main challenges for the autonomous wheel-legged robot due to its complex mechanical system. The wheel-legged robot has four legs and four wheels, and the wheels are installed on the end of the foot. To guarantee the tracking performance of the wheel-legged robot, effective approaches for reliable tracking control should be investigated with the consideration of the robot modeling, kinematics and dynamics constraints designing. In this paper, model predictive control based path following controller is designed and employed to improve the tracking performance for the wheel-legged robot BIT-NAZA. Experiments with the wheel-legged robot are performed to validate the performance of the proposed control strategy. The results demonstrate that the proposed methodology can achieve promising tracking performance in terms of accuracy.

Key Words: Model Predictive Control, Path Following, Wheel-legged Robot, Modeling

1 INTRODUCTION

Recently, wheel-legged robots have become prevalent in recent years, as an emerging robot system, provide considerable advantages in kinematic dexterity, environmental adaptability and motion precision [1–3]. However, wheel-legged robots have more actuators than the legged robot or wheeled robot. In this paper, an electric parallel wheel-legged robot named BIT-NAZA is introduced. It has four legs which are four inverted Stewart platforms and four wheels which are installed on the end of the foot. Every Stewart platform has six electric cylinders driven by six motors, and every wheel is driven by a motor. The steering angle range of each leg is $-60^\circ \sim 60^\circ$. Because of the novel mechanical structure, BIT-NAZA has abundant motion modes, such as leg mode, wheel mode and wheel-leg mode. Path following is the basis of autonomous motion of robots, the robot steers flexibly by controlling the turning angle of each leg in wheel-leg mode. Therefore, this study focuses on the tracking control of the wheel-legged robot BIT-NAZA with the consideration of the kinematics modeling and kinematics and dynamics constraints.

During the past few years, the path following problem has been studied by many researchers. A variety of approaches to this problem have been proposed in the literature, using a wide range of analytic tools. For example, in [4], PID controller with feed-forward for tight path following was presented. Since the PID control technique is very accustomed to ordinary engineers, they can easily exploit the proposed method for the path following without any heavy mathematical background. In [5], fuzzy PID controller was proposed for a car-like mobile robot, it has bet-

ter convergence rate in comparison with standard PID controller. In [6], human driving behavior based fuzzy control was designed to implement an accurate and efficient path following of a wheeled mobile robot, it included two fuzzy controllers: one was fuzzy position controller; the other was fuzzy orientation controller. In [7], two fuzzy logic controllers were designed to force underactuated ships to follow a desired path, despite of the presence of environmental disturbances induced by wave, wind and ocean current. Modified composite nonlinear feedback strategy was proposed to improve the transient performance and eliminate the steady-state errors in path-following control in [8]. Discrete feedback linearization method was presented in [9] for path following for wheeled robots. [10] addressed an adaptive fuzzy path following control scheme of a mobile robot with uncertainty of its center of mass. The proposed design scheme guaranteed that all signals in the closed-loop system are bounded, and the tracking error converge to origin. Adaptive dynamic surface control based path following controller was designed in [11] for cooperative path following of underactuated marine surface vehicles via fast learning. Robust H_∞ based path following controller was presented [12] for autonomous ground vehicles with delays and data dropouts. Minimum-time features were considered in [13] for the designing of path following controller, with the aim of combining performance and energy-oriented optimization of the vehicle motion. The stability of the tracking system of wheel-footed robot was considered in [14], and tracking expected yaw rate based sliding mode control (SMC) method was proposed.

In recent years, intelligent path tracking controller was widely studied. In [15], adaptive RBF neural network based path following controller was designed for single un-

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manned surface vessels in formation control. In [16], model predictive control based path following controller was presented for autonomous vehicle, and a C-based algorithm was implemented to solve the nonlinear optimization problem in real-time. In [17], a model predictive controller was designed for a tractor-trailer system to solve the path tracking problem. Nonlinear model predictive control was designed in [18] for constrained output path following. Model predictive control based path following controller has become prevalent because of its ability to cope the control problem with multi-outputs and constraints.

In this paper, model predictive control based path following controller is proposed for an autonomous wheel-legged robot BIT-NAZA, whose purpose is to control the robot to drive on reference path with small lateral error and small heading error. Robot kinematics model is designed, as well as the path following error model. Then, model predictive control strategy is implemented to cope the control problem with multi-outputs and constraint.

The rest of this paper is organized as following. The robot BIT-NAZA is introduced in Section 2. The robot kinematics model and path following error model are described in Section 3. The MPC based path following controller is presented in Section 4. Evaluation results are illustrated in Section 5, followed by the conclusion in Section 6.

2 CONFIGURATION OF BIT-NAZA

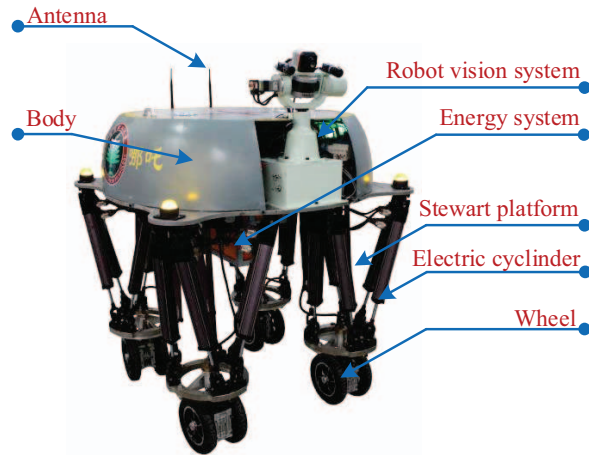


Figure 1: The wheel-legged robot BIT-NAZA

The wheel-legged robot BIT-NAZA is shown in Figure1. It includes four main subsystems: robot vision system, energy system, four legs and four wheels. Each leg of the robot is an inverted Stewart platform, and a wheel with the power 1200W is installed on the end of it. The Stewart is made up of six electric cylinders whose power are 400W. Stewart has six degrees of freedom, only the degree of freedom in horizontal rotation direction need be controlled when the robot is in the path following mode. The size of the robot is $1.4m \times 1.2m \times 1.2m$. The weight is 450Kg and the maximum load weight is about 300Kg.

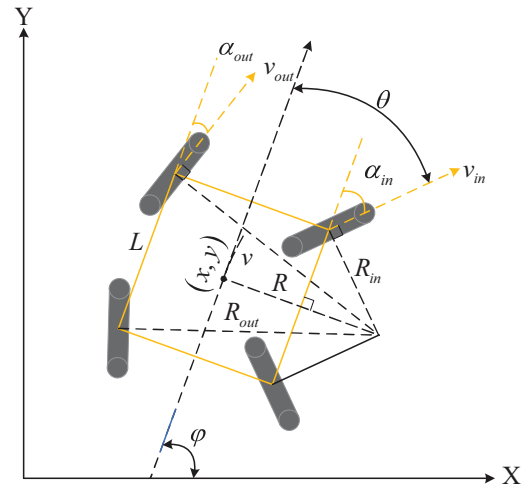


Figure 2: Steering kinematic model of the robot

3 PATH FOLLOWING PROBLEM

3.1 Robot Path Following Model

In order to design the path following controller of the wheel-legged robot, it is necessary to derive its kinematics model of path following. Steering kinematics of the robot is shown in Figure2. In Figure2, OXY is the geodetic coordinate system. (X_0, Y_0) is the coordinate at robot center. v_{in} is the inner wheel speed, v_{out} is the outer wheel speed. v is the speed of the robot. α_{in} is the steering angle of the inner wheels, α_{out} is the steering angle of the outer wheels. θ is defined as the steering angle of the robot, R is the steering radius. R_{in} is the radius of the circle of the inner wheel when the robot steers. R_{out} is the radius of the circle of the outer wheel when the robot steers. L is the wheel-track, and φ is the yaw angle of the robot.

At the center of the robot, the speed along the X-axis is

$$v_x = v \cos \varphi. \quad (1)$$

The speed along the Y-axis is

$$v_y = v \sin \varphi. \quad (2)$$

From Figure2, the yaw rate of the robot can be described as

$$\omega_r = \frac{2v \tan \theta}{L(1 + \tan \theta)}. \quad (3)$$

Kinematics model of the robot can be derived by combining (1), (2) and (3)

$$\dot{\chi} = f(\chi, u) = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\varphi} \end{bmatrix} = \begin{bmatrix} \cos \varphi \\ \sin \varphi \\ \frac{2 \tan \theta}{L(1 + \tan \theta)} \end{bmatrix} v \quad (4)$$

where $u(v, \theta)$ is the control output of the robot, $\chi(x, y, \varphi)$ is the state of the robot.

3.2 Path Following Error Model

Since the points on the given reference trajectory satisfy the kinematic equation (4), subscript r represents the reference value, we can get

$$\dot{\chi}_r = f(\chi_r, \mathbf{u}_r) \quad (5)$$

where $\chi_r = (x_r, y_r, \varphi_r)^T$ represents state value of the reference point. $\mathbf{u}_r = (v_r, \theta_r)$ is the output value of the reference point.

Taylor series expansion is used for $\dot{\chi} = f(\chi, \mathbf{u})$ at the reference point and higher order terms are ignored, it can be obtained

$$\dot{\chi} \approx f(\chi_r, \mathbf{u}_r) + \frac{\partial f(\chi, \mathbf{u})}{\partial \chi} \bigg|_{\chi=\chi_r, \mathbf{u}=\mathbf{u}_r} (\chi - \chi_r) + \frac{\partial f(\chi, \mathbf{u})}{\partial \mathbf{u}} \bigg|_{\chi=\chi_r, \mathbf{u}=\mathbf{u}_r} (\mathbf{u} - \mathbf{u}_r) \quad (6)$$

By subtracting equation (6) from equation (5), the path following error model of the wheel-legged robot can be obtained

$$\begin{aligned} \dot{\chi} &= \begin{bmatrix} \dot{x} - \dot{x}_r \\ \dot{y} - \dot{y}_r \\ \dot{\varphi} - \dot{\varphi}_r \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & -v_r(t) \sin \varphi_r(t) \\ 0 & 0 & v_r(t) \cos \varphi_r(t) \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x - x_r \\ y - y_r \\ \varphi - \varphi_r \end{bmatrix} + \\ &\quad \begin{bmatrix} \cos \varphi_r(t) & 0 \\ \sin \varphi_r(t) & 0 \\ \frac{2 \tan \theta_r(t)}{L(1+\tan \theta_r(t))} & \frac{2v_r(t)}{L(1+\tan \theta_r(t))^2(\cos \theta_r(t))^2} \end{bmatrix} \begin{bmatrix} v - v_r \\ \theta - \theta_r \end{bmatrix} \end{aligned} \quad (7)$$

(7) is discretized as

$$\tilde{\chi}(k+1) = \mathbf{A}\tilde{\chi}(k) + \mathbf{B}\tilde{\mathbf{u}}(k) \quad (8)$$

$$\text{where } \mathbf{A} = \begin{bmatrix} 1 & 0 & -v_r(t) \sin \varphi_r(t) T \\ 0 & 1 & v_r(t) \cos \varphi_r(t) T \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} \cos \varphi_r(t) T & 0 \\ \sin \varphi_r(t) T & 0 \\ \frac{2 \tan \theta_r(t) T}{L(1+\tan \theta_r(t))} & \frac{2v_r(t) T}{L(1+\tan \theta_r(t))^2(\cos \theta_r(t))^2} \end{bmatrix}$$

T is the sampling time.

4 MPC based PATH FOLLOWING CONTROLLER DESIGN

The design of objective function is an important part in the design of model predictive controller. It is mainly designed according to the requirements of the control system for the control index. For the path following system of the robot, the design purpose of objective function is to make the robot track the reference trajectory quickly and stably. The objective function is designed as

$$J(k) = \sum_{j=1}^{N_p} \|\tilde{\chi}(k+j|k)\|_Q^2 + \sum_{i=1}^{N_c-1} \|\Delta \mathbf{U}(k+i|k)\|_R^2 \quad (9)$$

where N_p is the predictive horizon, N_c is the control horizon, $\Delta \mathbf{U}$ is the control increment matrix.

We define

$$\xi(k|k) = \begin{bmatrix} \tilde{\chi}(k|k) \\ \tilde{\mathbf{u}}(k-1|k) \end{bmatrix}. \quad (10)$$

Then, (8) can be transformed as

$$\xi(k+1|k) = \tilde{\mathbf{A}}\xi(k|k) + \tilde{\mathbf{B}}\Delta \mathbf{U}(k|k) \quad (11)$$

where $\tilde{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0}_{m \times n} & \mathbf{I}_m \end{bmatrix}$, $\tilde{\mathbf{B}} = \begin{bmatrix} \mathbf{B} \\ \mathbf{I}_m \end{bmatrix}$, n is the dimension of state variables, m is the dimension of control variables.

The system prediction output is

$$\mathbf{Y}(k) = \psi \xi(k|k) + \Theta \Delta \mathbf{U}(k) \quad (12)$$

where

$$\mathbf{Y}(k) = \begin{bmatrix} \tilde{\chi}(k+1|k) \\ \tilde{\chi}(k+2|k) \\ \vdots \\ \tilde{\chi}(k+N_c|k) \\ \vdots \\ \tilde{\chi}(k+N_p|k) \end{bmatrix} \quad \psi = \begin{bmatrix} \tilde{\mathbf{A}} \\ \tilde{\mathbf{A}}^2 \\ \vdots \\ \tilde{\mathbf{A}}^{N_c} \\ \vdots \\ \tilde{\mathbf{A}}^{N_p} \end{bmatrix}$$

$$\Theta = \begin{bmatrix} \tilde{\mathbf{B}} & 0 & 0 & 0 \\ \tilde{\mathbf{A}}\tilde{\mathbf{B}} & \tilde{\mathbf{B}} & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{\mathbf{A}}^{N_c-1}\tilde{\mathbf{B}} & \tilde{\mathbf{A}}^{N_c-2}\tilde{\mathbf{B}} & \cdots & \tilde{\mathbf{B}} \\ \tilde{\mathbf{A}}^{N_c}\tilde{\mathbf{B}} & \tilde{\mathbf{A}}^{N_c-1}\tilde{\mathbf{B}} & \cdots & \tilde{\mathbf{A}}\tilde{\mathbf{B}} \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{\mathbf{A}}^{N_p-1}\tilde{\mathbf{B}} & \tilde{\mathbf{A}}^{N_p-2}\tilde{\mathbf{B}} & \cdots & \tilde{\mathbf{A}}^{N_p-N_c-1}\tilde{\mathbf{B}} \end{bmatrix}$$

$$\Delta \mathbf{U}(k) = \begin{bmatrix} \Delta u(k|k) \\ \Delta u(k+1|k) \\ \vdots \\ \Delta u(k+N_c|k) \end{bmatrix}.$$

MPC has the ability to deal with the constraints of control variables in the system. For the wheel-legged robot, the constraints of control variables include the steering angle, steering angle rate, longitudinal speed, longitudinal acceleration, etc. The constraints of control variables can be described as

$$\mathbf{u}_{\min} \leq \mathbf{u} \leq \mathbf{u}_{\max}, \quad k = 0, 1, \dots, N_c - 1 \quad (13)$$

The constraints of control increment variables can be described as

$$\Delta \mathbf{u}_{\min} \leq \Delta \mathbf{u} \leq \Delta \mathbf{u}_{\max}, \quad k = 0, 1, \dots, N_c - 1 \quad (14)$$

Then, it can be obtained

$$\mathbf{u}(k) = \mathbf{u}(k-1) + \Delta \mathbf{u}(k) \quad (15)$$

We set

$$\mathbf{U}_k = \mathbf{1}_{N_c} \otimes \mathbf{u}(k-1) \quad (16)$$

$$\mathbf{F} = \underbrace{\begin{bmatrix} 1 & 0 & \cdots & \cdots & 0 \\ 1 & 1 & 0 & \cdots & 0 \\ 1 & 1 & 1 & \ddots & 0 \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 1 & 1 & \cdots & 1 & 1 \end{bmatrix}}_{N_c \times N_c} \otimes \mathbf{I}_m \quad (17)$$

where $\mathbf{1}_{N_c}$ is the column vector with N_c rows, $\mathbf{I}_m$ is unit matrix with dimension m , $\otimes$ is Kronecker product, $\mathbf{u}(k-1)$ is the control value of the robot at the last moment.

By combining (15) and (17), (13) can be transformed as

$$\mathbf{U}_{\min} \leq \mathbf{F}\Delta\mathbf{U}_k + \mathbf{U}_k \leq \mathbf{U}_{\max} \quad (18)$$

where $\mathbf{U}_{\min}$, $\mathbf{U}_{\max}$ are the set of minimum and maximum control values.

Then, the path following objective function of the robot is transformed into the standard quadratic form with constraints, and the optimization problems are described as

$$\begin{aligned} J(\xi(k), \mathbf{u}(k-1), \Delta\mathbf{U}(k)) \\ = \Delta\mathbf{U}(k)^T \mathbf{H} \Delta\mathbf{U}(k) + \Delta\mathbf{U}(k)^T \mathbf{G} \end{aligned} \quad (19)$$

$$s.t. \quad \Delta\mathbf{U}_{\min} \leq \Delta\mathbf{U}_k \leq \Delta\mathbf{U}_{\max}$$

$$\mathbf{U}_{\min} \leq \mathbf{F}\Delta\mathbf{U}_k + \mathbf{U}_k \leq \mathbf{U}_{\max}$$

where $\mathbf{H} = \Theta^T \mathbf{Q} \Theta + \mathbf{R}$, $\mathbf{G} = 2\Theta^T \mathbf{Q} \mathbf{e}$. $\mathbf{e}$ is the tracking error of the robot in predictive horizon.

The control output increment in predictive horizon can be obtained by solving (19) in every control cycle

$$\Delta\mathbf{U}_k^* = [\Delta\mathbf{u}_k^*, \Delta\mathbf{u}_{k+1}^*, \dots, \Delta\mathbf{u}_{k+N_c-1}^*]^T \quad (20)$$

Take the first element of the control sequence as the actual input of the robot path following system

$$\mathbf{u}(k) = \mathbf{u}(k-1) + \Delta\mathbf{u}_k^* \quad (21)$$

In order to make the acceleration, deceleration and steering of the robot more stable, the constraint condition of the control value is designed as

constraints of control values:

$$\begin{bmatrix} -0.2 \\ -30 \end{bmatrix} \leq \mathbf{u} \leq \begin{bmatrix} 0.2 \\ 30 \end{bmatrix},$$

constraints of control incremental values:

$$\begin{bmatrix} -0.01 \\ -0.05 \end{bmatrix} \leq \Delta\mathbf{u} \leq \begin{bmatrix} 0.01 \\ 0.05 \end{bmatrix}.$$

5 CONTROLLER EVALUATION

5.1 Adams-Matlab Co-Simulation Platform

In order to verify the path following control effect of the proposed algorithm, a series of simulation experiments are designed. The simulation environment is a co-simulation system which includes Adams and Matlab/Simulink. Mechanical structure of the wheel-legged robot, setting of mechanical constraints and setting of actuator constraints are implemented in Adams environment. Robot kinematics

and dynamics can be simulated in Adams. Because of Matlab provide rich library functions for online optimization, digital processing, equation solving, etc. Thus, complex mathematical operations are handled in Matlab. In addition, Simulink provides a visual interface. The overall control system of the robot is built in Simulink environment. The visual graphic editing interface makes the design of the modular distributed control system of the robot more convenient.

In order to verify the control performance of the designed path following controller, the simulation environment is designed as shown in Figure3. The robot needs to pass a 7m straight line, two arcs with a radius of 5m, and a 7m straight line. There are narrow channels at the junction of different paths. The width of the narrow channels is only 20cm wider than that of the robot. If the robot path following control accuracy is high enough, the robot will not touch the channel wall when passing through the narrow channel, on the contrary, the robot will collide with the wall and increase the lateral deviation.

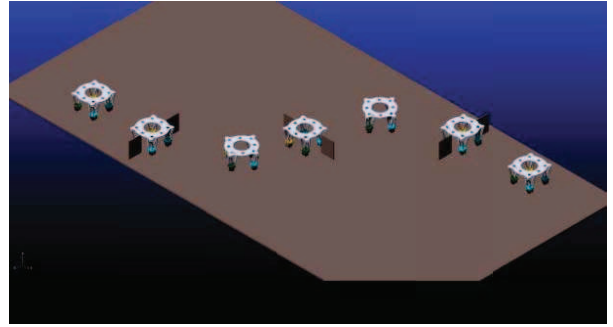


Figure 3: The designed simulation environment

Schematic diagram of the simulation environment is shown in Figure4. In the simulation environment, the current coordinates, speed, course angle and other information of the wheel-legged robot can be transmitted into Simulink through the interface with Adams. According to the path following error model, the lateral error and course angle error are calculated. Then, the robot is controlled to follow the reference trajectory.

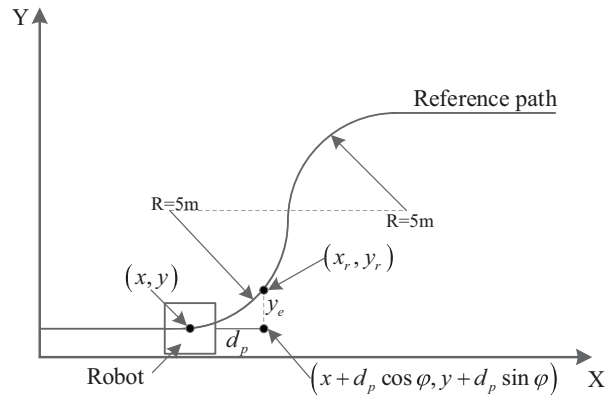
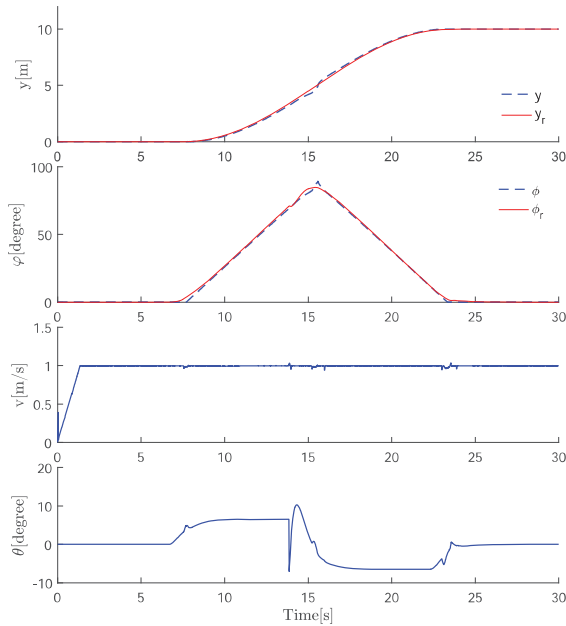
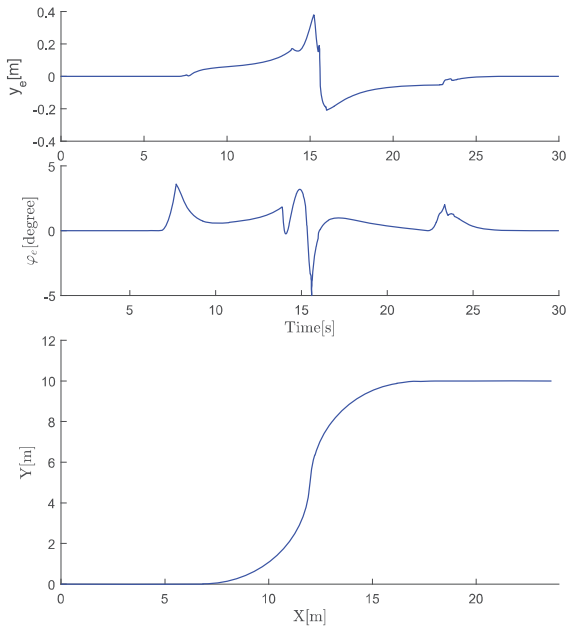


Figure 4: Schematic diagram of the simulation environment

5.2 Simulation Results



(a) Reference and actual curve

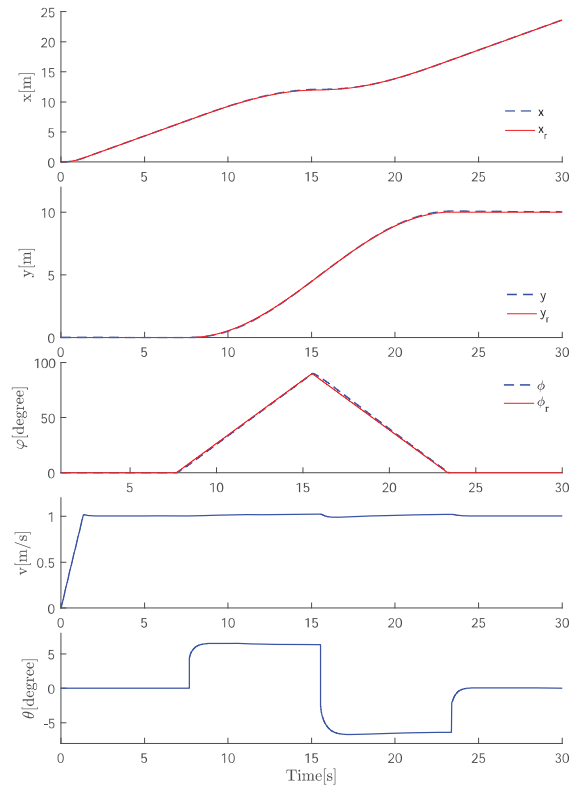


(b) Lateral and course angle error curve, actual path following curve

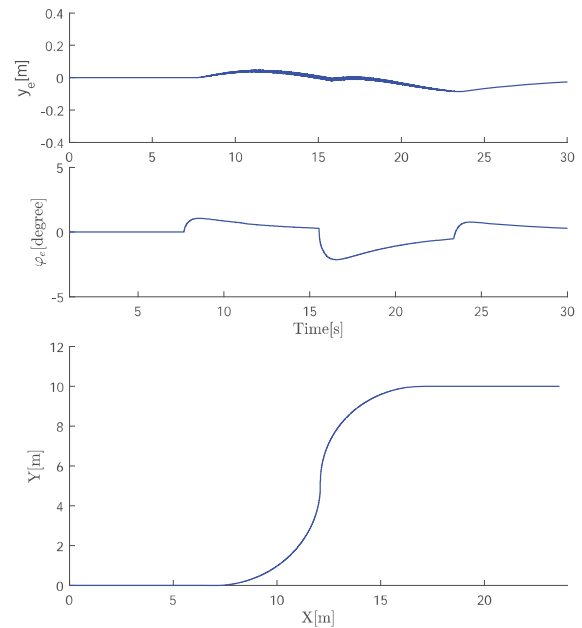
Figure 5: SMC based path following controller

In order to verify the control performance of the proposed MPC based path following controller, sliding mode control based path following controller proposed in [14] is used for comparison. Figure5 shows the control performance of SMC based path following controller. Figure6 shows the control performance of MPC based path following controller. The reference path is shown in Figure4. The speed of the robot is 1m/s.

In Figure5(a), from the top to the bottom, there are the comparison between the horizontal position of the reference path and the actual horizontal position of the robot,



(a) Reference and actual curve



(b) Lateral and course angle error curve, actual path following curve

Figure 6: MPC based path following controller

the comparison between the course angle of the reference path and the actual course angle of the robot, the speed of the robot, and the steering angle of the robot. In Figure 5(b), from top to bottom, there are robot lateral error, robot course angle error and actual trajectory of the robot. From the lateral error curve and course angle error curve, it can be seen the path following error of the robot is small when it is driving on the straight line part. The lateral error and course angle error gradually increase when the robot enters the first circular arc part. When the robot threads through the second narrow channel, the lateral error reaches 0.15m which is bigger than the safe distance. Therefore, the robot body is slightly in contact with the wall, and the path following error of the robot increase instantaneously. According to the numerical experiment results, as a typical mature controller in modern control theory, the sliding mode controller has the advantages of convenient engineering implementation and high precision, but it is not suitable for application in some occasions with high precision requirements.

Figure 6(a) shows the horizontal position, course angle, speed and steering angle of the robot. Figure 6(b) shows the lateral error, course angle error and actual trajectory of the robot. From Figure 6(b), it is obvious that the lateral error of the robot is less than 0.1m when passing through the narrow channel. Therefore, there is no contact between the robot body and the narrow channel wall. The speed and steering angle of the robot are within the constraints. The tracking accuracy is better than that of SMC based path following controller.

6 CONCLUSION

The problem of path following control of an autonomous wheel-legged robot BIT-NAZA is addressed in this paper. The wheel-legged robot BIT-NAZA is introduced. It has flexible motion modes benefit from the special wheel and leg combination mode. However, due to the complex mechanical structure, the motion control is still the main challenge for the autonomous wheel-legged robot, especially the path following control. To this end, we designed a MPC based path following controller for the wheel-legged robot. The kinematics model of the robot is designed, as well as the path following error model. Then, model predictive control strategy is implemented to cope the control problem with multi-outputs and constraints. Simulation results with the wheel-legged robot are performed to validate the performance of the proposed control strategy, and the results are compared with that of the SMC based path following controller. The results demonstrate that the proposed methodology can achieve promising tracking performance in terms of accuracy.

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